

Air University



Computer Networks (Lab)

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Assignment # 01

TASK 1: Device Purposes and Interfaces

1. Hub

Main Purpose	Common Interface
A hub operates at Layer 1 (Physical Layer) and sends incoming data frames to all connected ports simultaneously. It doesn't process MAC or IP addresses, creating a single collision domain. Hubs are mainly found in small, older networks and are generally seen as outdated in today's networking landscape.	Fast Ethernet ports (such as Fa0/0, Fa0/1, etc.) or standard Ethernet ports are connected using Copper Straight-Through cables.

2. Switch

Main Purpose	Common Interface
A Layer 2 switch intelligently forwards frames using MAC addresses, sending them only to the specific port where the target device resides. This approach minimizes unnecessary broadcast traffic and collisions, greatly enhancing network efficiency compared to hubs.	Fast Ethernet (Fa0/0, Fa0/1, etc.) or Gigabit Ethernet ports (Gi0/1, Gi0/2, etc.), used for connections to PCs, routers, and other switches.

3. Router

Main Purpose	Common Interface
Operating at Layer 3, a router directs packets between different networks based on IP addresses. It establishes logical boundaries, assigns IP subnets, and facilitates communication between Local Area Networks (LANs) and Wide Area Networks (WANs).	Fast Ethernet (Fa0/0, Fa0/1) for LAN connections and Serial ports (Serial0/0, Serial0/1) for WAN connections.

4. Wireless Access Point (AP)

Main Purpose	Common Interface
An access point expands wired network coverage wirelessly, allowing devices to connect via Wi-Fi. It connects to a switch or router through a wired Ethernet connection and transmits wireless signals based on 802.11 standards (Wi-Fi).	Fast Ethernet or Ethernet port (Port 0, Port 1) for the uplink to a switch, along with a wireless radio module for Wi-Fi transmission.

TASK 2: CORRECT CABLE TYPES

Connection Type	Correct Cable Type	Reasoning
PC ↔ Switch	Copper Straight-Through	This is a DTE-to-DCE connection and the standard for connecting unlike devices.
Switch ↔ Router (Ethernet)	Copper Straight-Through	It's a DTE-to-DCE connection, linking different devices with varied pin configurations.
Switch ↔ Switch (Ethernet)	Copper Cross-Over	Since both devices are of the same type, it requires a Tx/Rx swap to link them successfully.
Router ↔ Router (WAN serial link)	Serial DCE / Serial DTE	WAN connections utilize serial interfaces; the DCE/DTE roles are crucial for clock signal generation.
PC ↔ Router (for initial CLI setup)	Console (Rollover)	This out-of-band management method allows access when the device lacks a network configuration.
Switch ↔ Switch (long distance)	Fiber	Ideal for long-distance, high-bandwidth connections that require the use of fiber optics.

TASK 3: BRIEF ANSWERS**1. Difference Between Console Cable and Ethernet Cable**

Console (Rollover) Cable	Ethernet Cable (Straight-Through/Cross-Over)
This cable is used for out-of-band management, linking a computer to a device's console port (serial/management port) for initial configuration and CLI access when the device doesn't have an IP address or network connectivity.	Utilized for data transmission and network communication among devices. It operates at Layer 2/3 and must ensure correct pin alignment based on the types of connecting devices.

2. Green Link Light vs. Amber/Orange Link Light on a Switch Port

Green Light	Amber/Orange Light
Signals an active, stable connection operating at optimal speed (usually 1 Gbps for gigabit ports). This means the link is well-established and data can flow freely.	Indicates a connection at a reduced speed (either 100 Mbps or 10 Mbps) or suggests potential issues, such as Spanning Tree Protocol (STP) blocking, mismatched speed/duplex settings, or faulty cables. While traffic may still flow, it's at a lower speed.

THE END
